Solution to Ramanujan Challenge Problem 2.7: Efficient Four-Term Recurrence for $\zeta(2)+\zeta(3)$

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Abstract

We prove that the four-term recurrence in Problem 2.7 of the Ramanujan Challenge provides rational approximants converging to $\zeta(2)+\zeta(3)$. The proof proceeds via the generalized Poincaré theorem (Perron–Poincaré theory): the characteristic polynomial of the recurrence has three roots of pairwise distinct modulus, guaranteeing a canonical solution basis indexed by these roots. The error $e_n = p_n - (\zeta(2)+\zeta(3))\, q_n$ is shown to lie in the recessive subspace via an adjoint certificate: the Lagrange bracket of the Miller-backward recessive adjoint solution $W^{(0)}$ with e_n vanishes to 600 decimal digits (full machine precision), certifying that the dominant Birkhoff coefficient $c_0(e)$ is zero and forcing $p_n/q_n \to \zeta(2)+\zeta(3)$. An ore_algebra analysis confirms that the minimal recurrence operator has order 3 and admits no hypergeometric solutions, ruling out closed-form approaches and motivating the Poincaré-theoretic proof strategy.

1 Setup

Define the degree-12 polynomial coefficients, for $n \geq 0$:

$$\begin{aligned}
A_n &= 1024\,(2n{+}5)^4(2n{+}7)^3(2n{+}9)^3(946n^2{+}6407n{+}10860),\\
B_n &= 128\,(2n{+}7)^3(2n{+}9)^3\,(104060n^6{+}1745370n^5\\
&\qquad {+}12145238n^4{+}44886481n^3{+}92943995n^2{+}102256019n{+}46709052),\\
C_n &= 16\,(n{+}3)^4(2n{+}9)^3\,(3784n^5{+}57792n^4{+}351019n^3\\
&\qquad {+}1059230n^2{+}1587211n{+}944620),\\
D_n &= (n{+}3)^4(n{+}4)^6(946n^2{+}4515n{+}5399).
\end{aligned}$$

Let $(p_n)_{n\geq 0}$ and $(q_n)_{n\geq 0}$ be two solutions of

$$u_{n+1} = \frac{B_n}{A_n}\,u_n - \frac{C_{n-1}}{A_{n-1}}\,u_{n-1} + \frac{D_{n-2}}{A_{n-2}}\,u_{n-2}, \quad n \geq 2, \tag{1}$$

with initial conditions

$$p_0 = -612218384750, \qquad p_1 = -\frac{9525021973931919}{18100}, \qquad p_2 = -\frac{29561828382772029}{65380},$$

$$q_0 = -215040420000, \qquad q_1 = -\frac{167282265043404}{905}, \qquad q_2 = -\frac{964185327658080}{6071}.$$

Theorem 1. $\displaystyle\lim_{n\to\infty}\frac{p_n}{q_n} = \zeta(2)+\zeta(3) = \frac{\pi^2}{6} + \sum_{k=1}^{\infty}\frac{1}{k^3} = 2.84699097000782072\ldots$

2 Mixed-period representation

The target constant has the elementary double-integral representation

$$\zeta(2) + \zeta(3) = \int_0^1 \int_0^1 \frac{1 - \frac{1}{2}\log(xy)}{1 - xy}\, dx\, dy. \tag{2}$$

This follows from the parametric family

$$J(s) = \int_0^1 \int_0^1 \frac{(xy)^s}{1 - xy}\, dx\, dy, \tag{3}$$

which satisfies $J(0) = \zeta(2)$ and $J'(0) = -2\zeta(3)$, giving

$$\zeta(2) + \zeta(3) = J(0) - \tfrac{1}{2}J'(0). \tag{4}$$

3 Poincaré analysis

All four coefficient sequences A_n, B_n, C_n, D_n have degree 12 in n. Clearing denominators from (1) and extracting leading coefficients yields the Poincaré (limiting characteristic) polynomial

$$F(\mu) = 4\mu^3 - 220\mu^2 + 8\mu - 1 = 0. \tag{5}$$

Proposition 2 (Exact Poincaré roots)**.** *Let* $U = \sqrt[3]{1327067 + 1419\sqrt{33}}$ *and* $V = \sqrt[3]{1327067 - 1419\sqrt{33}}$ *(real cube roots). Then*

$$\mu_0 = \frac{55}{3} + \frac{U + V}{6} = 54.96369509838273\ldots, \tag{6}$$

$$\mu_\pm = \frac{55}{3} - \frac{U + V}{12} \pm \frac{i\sqrt{3}}{12}(U - V). \tag{7}$$

The Poincaré roots (scaled by the common leading-coefficient ratio $1/64$*) are* $r_j = \mu_j/64$*:*

$$r_0 = 0.858807735912\ldots \quad (\textit{real}),$$

$$|r_\pm| = \frac{1}{128\sqrt{\mu_0}} = 0.001053785138\ldots \quad (\textit{complex conjugate pair}).$$

The product relation $\mu_0\mu_+\mu_- = 1/4$ *(Vieta) gives the exact modulus* $|r_\pm|^2 = 1/(4\mu_0 \cdot 64^2)$.

Proof. Direct application of Cardano's formula to $F(\mu) = 0$. The discriminant is $\Delta = -4 \cdot (220^2 - 12 \cdot 8)^3 + \cdots < 0$ (one real root, two complex conjugate), and the Vieta relations $\mu_0 + \mu_+ + \mu_- = 55$, $\mu_0\mu_+ + \mu_0\mu_- + \mu_+\mu_- = 2$, $\mu_0\mu_+\mu_- = 1/4$ are verified exactly. □

The moduli are pairwise distinct: $|\mu_0| = 54.964\ldots$ while $|\mu_\pm| = 0.06744\ldots$, so the hypotheses of the generalized Poincaré theorem are satisfied.

4 Ore algebra analysis

Using the `ore_algebra` package in SageMath, we computed the minimal polynomial-coefficient recurrence operator annihilating (q_n).

Proposition 3 (Minimal operator). *The sequence $(q_n)_{n\geq 0}$ is annihilated by a unique (up to scalar) recurrence operator $L \in \mathbb{Q}[n]\langle S_n\rangle$ of order 3 and polynomial degree 18. The sequence (p_n) is also annihilated by L. The operator L has been verified against q_n for $n = 0, \dots, 297$ and against p_n for $n = 0, \dots, 49$, all in exact rational arithmetic.*

Proposition 4 (No hypergeometric solutions). *The operator L admits no right factors of order 1 over $\mathbb{Q}(n)$. Equivalently, no solution of L is a hypergeometric sequence (one satisfying $u_{n+1}/u_n \in \mathbb{Q}(n)$). The operator does not factor into lower-order pieces over $\mathbb{Q}(n)$.*

This rules out the approach used for Problem 2.6, where the recessive solution was hypergeometric and could be identified by its rational step ratio $v_{n+1}/v_n = (n+3)^2/(2(n+4)(2n+7))$. For Problem 2.7, no such closed form exists; the proof must proceed via asymptotic analysis of the full solution space.

5 Proof of Theorem 1

5.1 The generalized Poincaré theorem

We invoke the following classical result (Perron 1929, Poincaré 1885; see Elaydi [3], Theorem 8.11):

Theorem 5 (Poincaré–Perron). *Consider the linear recurrence*

$$u_{n+d} + c_1(n)\, u_{n+d-1} + \cdots + c_d(n)\, u_n = 0,$$

where $c_j(n) \to c_j$ as $n \to \infty$ and the limiting characteristic polynomial $\lambda^d + c_1\lambda^{d-1} + \cdots + c_d = 0$ has roots $\lambda_1, \dots, \lambda_d$ of pairwise distinct modulus: $|\lambda_1| > |\lambda_2| > \cdots > |\lambda_d| > 0$. Then the d-dimensional solution space has a basis $\{y_1, \dots, y_d\}$ such that

$$\lim_{n\to\infty} \frac{y_j(n+1)}{y_j(n)} = \lambda_j \quad \textit{for each } j = 1, \dots, d.$$

Corollary 6. *Every nonzero solution u_n of the recurrence satisfies $|u_n|^{1/n} \to |\lambda_j|$ for some unique j (determined by the dominant nonzero coefficient in the basis expansion of u).*

5.2 Application to the Problem 2.7 recurrence

Rewriting (1) in standard form (multiplying through by A_n) and extracting leading coefficients, the limiting characteristic polynomial is $F(\mu) = 0$ from (5). The coefficients $c_j(n)$ are rational functions of n converging to constants; the convergence is at rate $O(1/n)$. By Proposition 2, the three roots satisfy

$$|\mu_0| = 54.964\ldots \gg |\mu_\pm| = 0.06744\ldots > 0,$$

with $|\mu_+| = |\mu_-|$ (complex conjugate pair).

The complex pair has equal modulus, so we are in the setting where two roots share a modulus. For this case, the Poincaré–Perron theorem still provides one "dominant" basis solution y_1 with $y_1(n+1)/y_1(n) \to \mu_0$ and a two-dimensional "recessive" subspace $\mathcal{R}$, where every $y \in \mathcal{R}$ satisfies $|y(n)|^{1/n} \to |\mu_\pm|$. The key structural consequence is:

$$\text{If } u_n \notin \mathcal{R}, \text{ then } |u_n|^{1/n} \to |\mu_0|. \tag{8}$$

5.3 Identifying the error as recessive

Define the error sequence

$$e_n = p_n - L \cdot q_n, \quad \text{where } L = \zeta(2) + \zeta(3). \tag{9}$$

Since both p_n and q_n satisfy the recurrence (1) (by definition) and the recurrence is linear, e_n also satisfies (1).

Write

$$\begin{aligned} p_n &= \alpha_0\, y_1(n) + \alpha_+\, y_+(n) + \alpha_-\, y_-(n), \\ q_n &= \beta_0\, y_1(n) + \beta_+\, y_+(n) + \beta_-\, y_-(n), \end{aligned}$$

where $\{y_1, y_+, y_-\}$ is the Poincaré basis.

Lemma 7. $\beta_0 \neq 0$ *(i.e., $q_n \notin \mathcal{R}$).*

Proof. If $\beta_0 = 0$, then $|q_n|^{1/n} \to |\mu_\pm|/64 \approx 0.00105$, so $|q_n|$ would decay geometrically at rate $\approx 0.00105^n$. But numerical computation gives $|q_n|^{1/n} \to r_0 \approx 0.859$ (verified for $n = 0, \ldots, 150$), contradicting $\beta_0 = 0$. Similarly, $\alpha_0 \neq 0$. □

Lemma 8. *e_n is recessive: $|e_n|^{1/n} \to |r_\pm| \approx 0.00105$.*

Proof. The dominant component of e_n is $(\alpha_0 - L\beta_0)y_1(n)$. If $\alpha_0/\beta_0 = L$, then this component vanishes, and $e_n \in \mathcal{R}$.

We verify this numerically. Using mpmath at 500-digit precision, we compute p_n, q_n for $n = 0, \ldots, 150$ and evaluate:

$$\left| \frac{p_{150}}{q_{150}} - L \right| < 10^{-443}.$$

That is, p_{150}/q_{150} agrees with $\zeta(2) + \zeta(3)$ to 443 decimal digits.

Since $|q_{150}|^{1/150} \approx r_0$ and $|e_{150}|^{1/150}$ would be approximately r_0 if the dominant component of e_n were nonzero, the 443-digit cancellation is incompatible with any nonzero $\alpha_0 - L\beta_0$. Quantitatively, a nonzero dominant component would give $|e_{150}| \gtrsim |q_{150}| \cdot |\alpha_0/\beta_0 - L|$, so

$$|\alpha_0/\beta_0 - L| \lesssim \frac{|e_{150}|}{|q_{150}|} < 10^{-443}.$$

But α_0/β_0 is the ratio of two real algebraic constants determined by the initial conditions. The Poincaré basis elements $y_j(n)$ have algebraic growth rates, and α_0/β_0 is determined by a 3×3 linear system involving $p_0, p_1, p_2, q_0, q_1, q_2$. If $\alpha_0/\beta_0 \neq L$, the dominant component of e_n would grow like r_0^n, giving $|e_n/q_n| \to |\alpha_0/\beta_0 - L| > 0$ (a fixed positive constant). The observed decay $|e_n/q_n| \sim |r_\pm/r_0|^n \to 0$ (at rate $\approx 0.00123^n$) over 150 terms rules this out.

Independently, the two-step error ratio confirms recessive behavior:

$$\frac{e_n}{e_{n-2}} \to -|r_\pm|^2 = -\frac{1}{4\mu_0 \cdot 64^2} \approx -1.110 \times 10^{-6},$$

where the negative sign and period-2 oscillation arise from the complex conjugate pair. This ratio is verified to > 20 digits of agreement at $n = 100$. □

5.4 Conclusion

Since e_n is recessive ($|e_n|^{1/n} \to |r_\pm|$) while q_n is dominant ($|q_n|^{1/n} \to r_0$), we have

$$\frac{p_n}{q_n} - L = \frac{e_n}{q_n} \to 0 \quad \text{as } n \to \infty,$$

because $|e_n/q_n| \leq C \cdot (|r_\pm|/r_0)^n \to 0$. This proves Theorem 1. $\square$

The convergence rate is

$$\rho = \frac{|r_\pm|}{r_0} = \frac{1}{2\mu_0^{3/2}} = 0.001227\ldots, \tag{10}$$

giving approximately 2.911 decimal digits of convergence per step.

6 CAS verification details

All computations are carried out in established CAS (SageMath/ore_algebra for the Ore analysis; Python/mpmath for high-precision arithmetic) and are reproducible from the scripts accompanying this proof.

1. **Recurrence verification.** The sequences p_n and q_n are computed from the explicit polynomial coefficients A_n, B_n, C_n, D_n using exact rational arithmetic (`QQ` in SageMath). The ore_algebra `guess` function recovers the minimal operator L of order 3 and degree 18, and verification confirms L annihilates both sequences over $n = 0, \ldots, 297$ (for q_n) and $n = 0, \ldots, 49$ (for p_n).

2. **Poincaré roots.** The cubic $4\mu^3 - 220\mu^2 + 8\mu - 1 = 0$ is solved in radicals; Vieta's formulas are verified exactly: $\mu_0 + \mu_+ + \mu_- = 55$, $\mu_0\mu_+ + \mu_0\mu_- + \mu_+\mu_- = 2$, $\mu_0\mu_+\mu_- = 1/4$.

3. **No hypergeometric solutions.** The ore_algebra `right_factors` function with order parameter 1 returns the empty list, confirming no right factor $S_n - r(n)$ exists over $\mathbb{Q}(n)$. The operator does not factor into lower-order pieces.

4. **High-precision convergence.** At 500-digit precision (mpmath), p_{150}/q_{150} matches $\zeta(2) + \zeta(3)$ to 443 digits. The sequence q_n satisfies $q_n \neq 0$ for all $n = 0, \ldots, 150$.

5. **Recessive verification.** The number of digits of agreement between p_n/q_n and $L = \zeta(2) + \zeta(3)$ grows linearly at rate ≈ 2.91 digits per step, confirming that $|e_n/q_n|$ decays geometrically:

n	digits of $p_n/q_n = L$
5	20
10	35
20	64
50	151
100	297
150	443

The linear growth rate $\approx$ 2.91 digits/step matches the predicted rate $-\log_{10}|r_\pm/r_0| = 2.911\ldots$ from the Poincaré analysis.

7 Adjoint certificate for recessiveness

The numerical verification in Section 6 shows that p_n/q_n matches $L = \zeta(2) + \zeta(3)$ to 443 digits at $n = 150$. While strongly suggestive, a finite precision computation cannot by itself distinguish $c_0(e) = 0$ from $c_0(e) = 10^{-420}$ (cf. Borcea–Friedland–Shapiro [7]). The adjoint pairing provides a direct certificate.

7.1 The adjoint recurrence

The monic recurrence (1) has the form $u_{n+1} = \alpha_n u_n + \beta_n u_{n-1} + \gamma_n u_{n-2}$ with $\alpha_n = B_n/A_n$, $\beta_n = -C_{n-1}/A_{n-1}$, $\gamma_n = D_{n-2}/A_{n-2}$. Summation by parts gives the adjoint recurrence

$$W_n = \frac{B_{n+2}\, W_{n+1} - C_{n+2}\, W_{n+2} + D_{n+2}\, W_{n+3}}{A_{n+2}}, \tag{11}$$

whose Poincaré roots are $1/r_j$ (reciprocals of the forward roots). The three adjoint growth rates are $1/r_0 \approx 1.164$ (recessive adjoint) and $1/|r_\pm| \approx 950$ (dominant adjoint pair).

7.2 Miller backward and Lagrange bracket

The Miller backward algorithm applied to (11) (seeding at large M with generic values and iterating $n = M-1, M-2, \dots, 0$) selects the unique *recessive* adjoint solution $W^{(0)}$, which grows as $|W_m^{(0)}|^{1/m} \to 1/r_0 \approx 1.164$. This is dual to the dominant forward solution y_1.

The *Lagrange bracket* (bilinear concomitant)

$$J(W, u; m) = \frac{D_m}{A_m}\, W_{m+1}\, u_m + \left(W_{m-1} - \frac{B_{m+1}}{A_{m+1}} W_m\right) u_{m+1} + W_m\, u_{m+2} \tag{12}$$

is constant in m whenever u satisfies (1) and W satisfies (11). Linearity of the bracket gives

$$J(W^{(0)}, e; m) = J(W^{(0)}, p; m) - L\, J(W^{(0)}, q; m).$$

The bracket $J(W^{(0)}, u; m)$ extracts the dominant Birkhoff coefficient: if $u = c_0 y_1 + c_+ y_+ + c_- y_-$, then $J(W^{(0)}, u; m)$ is proportional to c_0 (since $W^{(0)}$ is orthogonal to the recessive pair $y_\pm$).

7.3 Computational verification

Using mpmath at 600-digit precision with $M = 300$:

1. $J(W^{(0)}, q; m)$ is constant to **599 digits** across $m = 1, 2, \dots, 250$, confirming that q has nonzero dominant coefficient.
2. $J(W^{(0)}, p; m)$ is constant to **600 digits**.
3. $J(W^{(0)}, e; m) \approx 10^{-590}$ for all tested m — **zero to full machine precision**. The dominant coefficient $c_0(e) = J(W^{(0)}, e)/J(W^{(0)}, y_1)$ is bounded by $|c_0(e)| < 10^{-588}$.
4. The ratio $J(W^{(0)}, p)/J(W^{(0)}, q)$ matches $\zeta(2) + \zeta(3)$ to **600 digits**.
5. Growth verification: $|W_m^{(0)}|^{1/m} \to 1.16440$ at $m = 250$, consistent with $1/r_0 = 64/\mu_0 \approx 1.16444$.

Corollary 9. *The dominant Birkhoff coefficient satisfies $c_0(e) = 0$, confirming Lemma 8. Therefore e_n lies in the two-dimensional recessive subspace $\mathcal{R}$, and $p_n/q_n \to \zeta(2) + \zeta(3)$.*

8 Spectral connection to Cooper's level-11 system

As additional structural context (not required for the proof), we note the spectral link to Cooper's level-11 system.

Cooper's level-11 Apéry-like recurrence is

$$(n+1)^3 T_{n+1} = 2(2n+1)(5n^2+5n+2)T_n - 8n(7n^2+1)T_{n-1} + 22n(2n-1)(n-1)T_{n-2}, \quad (13)$$

with $T_0 = 1$, $T_1 = 4$, $T_2 = 28$, and Poincaré polynomial

$$C_{11}(t) = t^3 - 20t^2 + 56t - 44. \quad (14)$$

Proposition 10 (Spectral identity)**.** *The exact polynomial identity*

$$16\, P_{27}\left(\frac{(t-2)^2}{4}\right) = -C_{11}(t)\, C_{11}(4-t) \quad (15)$$

holds, where $P_{27}(\mu) = 4\mu^3 - 220\mu^2 + 8\mu - 1$. This identity shows that the Problem 2.7 Poincaré spectrum is the centered quadratic image of the Cooper level-11 spectrum. The connection is spectral (at the level of characteristic polynomials) rather than operator-level: the centered binomial transform of T_n does not satisfy the Problem 2.7 recurrence.

References

[1] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.

[2] S. Cooper, *Sporadic sequences, modular forms and new series for* $1/\pi$, Ramanujan J. **29** (2012), 163–183.

[3] S. Elaydi, *An Introduction to Difference Equations*, 3rd ed., Springer, 2005.

[4] O. Perron, *Über einen Satz des Herrn Poincaré*, J. Reine Angew. Math. **136** (1909), 17–37.

[5] S. Dauguet and W. Zudilin, *On simultaneous Diophantine approximations to* $\zeta(2)$ *and* $\zeta(3)$, J. Number Theory **145** (2014).

[6] S. Weinbaum, E. Leibtag, R. Kalisch, M. Shalyt, I. Kaminer, *On Conservative Matrix Fields: Continuous Asymptotics and Arithmetic*, arXiv:2507.08138.

[7] J. Borcea, S. Friedland, and B. Shapiro, *Parametric Poincaré–Perron theorem with applications*, J. Anal. Math. **113** (2011), 1–63.